
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become the central lever for eliciting reasoning
2 in large language models, yet the mechanism by which RL reshapes a model
3 during training has remained poorly characterized. In this work, we resolve this
4 gap by explaining the RL training process through a unified account that couples
5 systematic empirical analysis with a rigorous theoretical model. Empirically, a
6 joint reasoning-pattern-level and token-level analysis of the training trajectory es-
7 tablishes a consistent picture: the intrinsic success rate of each individual reasoning
8 pattern stays essentially fixed, while RL concentrates its optimization on a sparse
9 subset of critical tokens and, through them, *reshapes the distribution over reasoning*
10 *patterns* to drive the observed changes in performance. Building on this evidence,
11 we develop a theoretical framework that captures the training dynamics of RL
12 under the two prevailing reward regimes: reinforcement learning with verifiable
13 rewards (RLVR) and reinforcement learning with internal feedback (RLIF). For
14 RLVR, we analyze two canonical regimes—one in which the model converges
15 rapidly to the optimal reasoning strategy, and one in which an entanglement stage
16 makes optimization difficult—and show that the base model’s reasoning quality is
17 the decisive factor governing convergence. For RLIF, we explain precisely why
18 internal rewards raise performance in the early phase of training and why continued
19 optimization can nonetheless drive the model below its starting point. Extensive
20 experiments across multiple RL methods, models, and mathematical reasoning
21 benchmarks corroborate every component of the analysis, jointly advancing the
22 theoretical understanding of RL post-training and yielding actionable guidance for
23 strengthening the reasoning capabilities of language models.

24 1 Introduction

25 State-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14] now
26 attain exceptional performance on demanding logical tasks spanning mathematics [30, 44, 41] and
27 programming [49, 40, 8]. Reinforcement learning (RL) is the key technique underpinning this
28 progress, repeatedly proving capable of elevating model reasoning to a qualitatively higher level.

29 This success has motivated a growing body of work seeking to explain the mechanisms of RL for
30 LLMs. Comparing pass@k performance, Yue et al. [42] observe that RL-post-trained models rarely
31 surpass their base models, suggesting that RL does not elicit fundamentally new reasoning patterns
32 but rather reweights existing ones. From an entropy standpoint, Cui et al. [9], Zhang et al. [45] prove
33 that RL-based methods reduce policy entropy, and Cui et al. [9] further establish an empirical link
34 between policy entropy and model performance. Complementing these results, Wang et al. [33], Huan
35 et al. [17] show that RL concentrates its updates on a sparse subset of critical tokens. Each of these
36 lines isolates one facet of the phenomenon; what remains missing—and what our work supplies—is

37 a unified account of the RL training *dynamics* that connects these facets and is grounded in both
38 empirical evidence and formal analysis.

39 To characterize the RL training process, we first conduct a systematic analysis that couples the
40 reasoning-pattern level with the token level across the entire training trajectory. Concretely, we
41 track next-token probability-rank shifts over training and, in parallel, apply both LLM-based and
42 rule-based procedures to extract and classify the reasoning patterns realized in model responses,
43 measuring how their success rates and selection frequencies evolve. This joint design yields clearer
44 and more decisive evidence than prior analyses [17, 42]: RL optimizes only a sparse subset of critical
45 tokens and, through them, *reshapes the distribution over reasoning patterns* to determine model
46 performance, while the intrinsic success rate of each individual pattern remains essentially invariant
47 throughout training. These two observations—sparse token-level intervention driving pattern-level
48 reallocation, atop fixed per-pattern success rates—are precisely the ingredients that motivate and
49 constrain the mathematical framework we develop next.

50 Guided by these findings, we develop a theoretical framework that formalizes reasoning as a two-stage
51 question–reason–answer process $q \rightarrow r \rightarrow a$: the model first selects a reasoning pattern conditioned
52 on the question, $\pi(r|q)$, and then generates a final answer conditioned on that pattern, $\pi(a|r, q)$, with
53 RL acting principally on the first, pattern-selection stage. Within this framework, we analyze the
54 training dynamics of RL under the two dominant reward types: verifiable rewards (RLVR) [14, 30]
55 and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove that optimization converges
56 to the reasoning pattern r^* with the highest success rate—exactly the behavior our experiments
57 exhibit—and we further delineate two distinct convergence regimes: models with strong initial
58 reasoning quality converge rapidly to the optimal pattern, whereas weaker models must first traverse
59 an entanglement stage in which a suboptimal pattern retards optimization. For RLIF, we explain why
60 internal rewards improve performance early in training and prove that continued optimization can
61 nonetheless converge to a state strictly worse than the base model, again matching what we observe
62 empirically. Dedicated case studies and simulations validate the theory across both regimes.

63 Our main contributions are as follows.

- 64 • **A unified empirical characterization of RL training dynamics.** We conduct a systematic analysis
65 that couples reasoning-pattern-level and token-level views across the full RL training trajectory.
66 Combining next-token probability-rank shifts with LLM-based and rule-based pattern analysis, we
67 deliver clearer and more decisive evidence than prior work [17, 42], establishing that RL optimizes
68 only a sparse subset of critical tokens and thereby *reshapes the distribution over reasoning patterns*
69 to determine performance, while the intrinsic success rate of each individual pattern remains
70 essentially invariant during training.
- 71 • **A theoretical framework with provable dynamics for both reward regimes.** We introduce a
72 formal two-stage model of reasoning, $q \rightarrow r \rightarrow a$ (question–reason–answer), and use it to analyze
73 the training dynamics of the two dominant RL paradigms. For RLVR, we prove convergence to the
74 highest-success-rate reasoning pattern and rigorously separate two regimes—rapid convergence
75 for strong base models versus a provably slow entanglement stage for weaker ones. For RLIF, we
76 prove that early-stage performance gains are guaranteed at initialization, yet continued training can
77 converge to a state strictly worse than the base model, thereby explaining both the benefits and the
78 failure mode of internal-feedback rewards.
- 79 • **Empirical validation and practical guidance.** We corroborate the framework through case studies
80 and numerical simulations that reproduce both RLVR regimes and RLIF’s non-monotone behavior.
81 In doing so, we close the gap between empirical observation and theoretical understanding of
82 RL training dynamics and distill concrete, reviewer-facing guidance for designing and improving
83 RL-based LLM post-training.

84 2 Related Works

85 **Reinforcement Learning for LLMs** Reinforcement learning has become a cornerstone technique for
86 advancing large language models (LLMs), both in aligning model behavior with human preferences
87 [25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. The reward model
88 sits at the center of this pipeline; classically it is trained on extensively curated, human-annotated
89 data [26, 25, 2]. A decisive recent shift replaces such costly supervision with rewards that are cheaply

obtainable and automatically verifiable, as in reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. This shift has, in turn, driven a wave of new optimization algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The empirical success of RL for LLMs has spurred extensive work on theoretical analysis and mechanism interpretation. Early efforts operated largely at a high level, studying how to design better rewards [28, 18, 38] and how to construct novel training algorithms [37, 11, 21]. As the field moved toward the simpler reward structures of RLVR and RLIF, attention turned to explaining *why* these paradigms are so effective, and several complementary accounts have emerged: Shao et al. [29] show that even spurious rewards can enhance reasoning by unlocking latent skills acquired during pretraining; Agarwal et al. [4] attribute the gains to entropy minimization; Gandhi et al. [13] isolate specific cognitive behaviors, such as backtracking, that improve reasoning; and Zhao et al. [46] demonstrate that RL fine-tuning amplifies pre-trained behaviors and converges toward a dominant output format, a phenomenon that Wang et al. [34], Cui et al. [10] further dissect through entropy collapse. Each of these perspectives illuminates a single symptom of RL training. In contrast, we analyze the RL training *dynamics* through the unifying lens of reasoning-pattern selection and, to our knowledge, are the first to support this view with a complete mathematical framework and accompanying theoretical guarantees.

3 Preliminaries

Reinforcement Learning for LLMs Let π_θ denote a language model with parameters θ , treated as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ autoregressively generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. RL optimizes the KL-regularized expected-reward objective

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} is the base reference policy and the hyperparameter β weights the KL-divergence penalty that constrains π_θ from drifting away from π_{ref} .

The reward $r_\phi(\mathbf{x}, \mathbf{y})$ admits several instantiations, ranging from a trained reward model [25] to a rule-based scoring function [30]. We concentrate on the two verification-light regimes that now dominate reasoning-oriented post-training: verifiable rewards (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

Under RLVR, the reward $r_\phi(\mathbf{x}, \mathbf{y})$ evaluates directly whether the generated answer $\mathbf{y}$ matches the ground-truth answer to $\mathbf{x}$, yielding the binary criterion

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

In contrast, RLIF [47, 4] dispenses with any external verifier and derives its reward entirely from intrinsic model signals. Following Zhao et al. [47], we take the RLIF reward to be the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution, so that higher reward corresponds to higher model confidence:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

Together, RLVR and RLIF span the two prevailing reward regimes for reasoning-oriented post-training, and our goal is to characterize the training dynamics of both—empirically and theoretically—within a single framework.

4 Experimental Exploration for RL Training

To characterize the RL training process, we conduct a systematic, multi-resolution study of RL under different reward types, organized so that each level of analysis explains the one above it. We begin with a high-level overview of the training procedure, which establishes a clear behavioral contrast: RLVR yields steady, monotone improvement, whereas RLIF is unstable and can ultimately

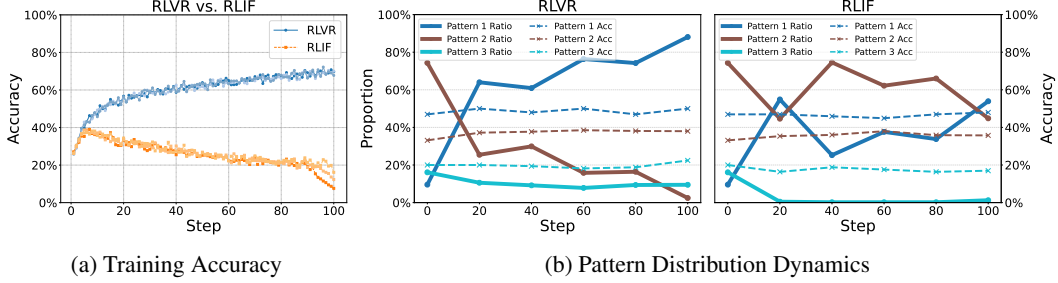


Figure 1: **(a)** Training trajectories of RLVR and RLIF on the MATH dataset (Qwen2.5-3B base model), with multiple runs per paradigm obtained by varying the random seed. The contrast is decisive: RLVR delivers stable, monotone gains, whereas RLIF drives accuracy up and then down. **(b)** The reasoning-pattern-level view of the same training. Solid lines give each pattern’s selection frequency among all patterns (left axis) and dotted lines give its corresponding accuracy (right axis). Under RLVR the model progressively concentrates on higher-accuracy patterns while abandoning lower-accuracy ones; under RLIF the pattern distribution evolves erratically and fails to specialize. Patterns are sorted and named by decreasing accuracy for readability, so pattern-selection shifts toward Pattern 1 correspond directly to gains in overall accuracy.

degrade performance below the base model. We then descend to a reasoning-pattern-level analysis, which localizes the cause of this contrast—RLVR progressively concentrates probability on reasoning patterns with higher success rates, while RLIF fails to specialize in effective patterns. Finally, a token-level analysis exposes the underlying mechanism: these pattern-level reallocations are driven by rank changes in a strikingly small fraction of tokens. Taken together, and in comparison with prior work [17, 42], these experiments furnish clearer and more decisive evidence that RL optimizes only a sparse subset of critical tokens and, through them, *reshapes the distribution over reasoning patterns* to determine model performance.

4.1 Training Procedure Overview for RLVR and RLIF

We first examine the complete training trajectories of RLVR and RLIF under a controlled comparison. We take Qwen2.5-3B [39] as the base model and train it on the MATH dataset [16], holding every other setting fixed across the two algorithms so that the sole source of variation is the reward.

As Figure 1a shows, the two methods trace sharply different performance trajectories. RLVR induces a stable optimization process in which accuracy improves steadily and converges smoothly over training. RLIF, by contrast, raises performance only transiently: after early gains, continued training drives accuracy down, in some runs below that of the original base model. This qualitative divergence—monotone improvement versus a rise-then-collapse trajectory—is exactly the phenomenon our subsequent fine-grained analysis is designed to explain.

4.2 Reasoning-pattern Level Analysis

To pinpoint the source of the performance disparity between RLVR and RLIF, we carry out a fine-grained analysis at the reasoning-pattern level. To define and extract patterns, we first collect responses from the base model and employ GPT-4o [19] to cluster them into distinct categories according to their keywords and logical structure, generating a descriptive summary for each category. Using this classifier, we then track both the selection frequency of each reasoning pattern and its associated success rate across the entire RL training procedure, so that changes in overall accuracy can be attributed to their pattern-level origin.

Concretely, we trace the joint dynamics of pattern selection and per-pattern accuracy under both RLVR and RLIF. Figure 1b reports this analysis on Number Theory tasks drawn from the MATH dataset [16], with consistent results across additional models and datasets deferred to Appendix A. Three findings emerge consistently:

- RLVR-trained models shift monotonically toward reasoning patterns with higher success rates, which directly accounts for the steady improvement in overall accuracy under RLVR.

- RLIF, in contrast, drives an unstable pattern distribution that never specializes in the more effective patterns, mirroring its erratic accuracy trajectory.
- Across both methods, the success rate of each individual reasoning pattern remains stable throughout training—so performance changes are attributable to *which* pattern is selected, not to any change within a pattern.

These results localize the RLVR–RLIF performance gap squarely in their divergent pattern-selection dynamics: RLVR reallocates probability toward high-success patterns while RLIF does not, even though the patterns themselves are equally effective under both. This pins the mechanism at the level of pattern selection and motivates the token-level analysis that follows, which asks what, within the model, actually implements these reallocations.

4.3 Token-level Analysis

To uncover the mechanism that implements the reasoning-pattern dynamics identified above, and building on Huan et al. [17], Yue et al. [42], we conduct a token-level analysis. We sample base-model responses to questions from GSM8K [6], MATH [16], and AIME [7], and then, holding these responses fixed, compare the rank each token receives under the base model against its rank under the RL-enhanced model, where a token’s rank is the position of its probability among all vocabulary tokens given the preceding context. As summarized in Table 1, the ranks are overwhelmingly stable: fewer than 10% of tokens change rank in essentially every task, method, and training step. RL therefore acts as a sparse intervention, adjusting the probabilities of a small set of critical tokens while leaving the bulk of the reasoning trajectory untouched.

Table 1: Fraction of tokens whose probability rank shifts after RL, by task, method, and training step. Across GSM8K, MATH, and AIME24, only a small minority of tokens change rank—below 10% in every case—confirming that RL intervenes sparsely at the token level.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

Taken together, the three levels of analysis converge on a single, clean characterization of RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

Beyond deepening the empirical understanding of RL post-training, this characterization supplies the two structural ingredients—sparse token-level control of pattern selection, atop fixed per-pattern success rates—on which we build the mathematical framework and formal analysis of the next section.

5 Theoretical Characterization of RL Training Dynamics

We now move from empirical characterization to formal explanation of the RL training process. Building directly on the two structural findings of the previous section, we abstract reasoning as a two-stage procedure—first selecting a reasoning pattern, then deducing an answer from it—in which RL principally optimizes the selection stage. This framework converts our empirical observations into provable statements about RL under both reward types. For RLVR, we analyze the training dynamics and rigorously separate two regimes with qualitatively distinct optimization behavior. For RLIF, we prove why internal rewards improve performance at the outset of training and, at the same time, why continued optimization can converge to a model strictly worse than the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize reasoning as the following two-stage generative process: given a question q , the model (1) selects a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that pattern. Pattern r_i is chosen with probability $p(r_i|q)$, and each pattern carries its own success rate—the probability that it leads to the correct answer. This decomposition lets us recast the RLIF

214 objective (Eq. 3.3) directly in terms of the pattern and answer distributions¹:

$$r_\phi(\{\mathbf{r}, \mathbf{a}\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(\mathbf{a}_j|q, \mathbf{r})) \right) \quad (5.1)$$

215 **Policy Parameterization.** Let $\mathcal{V} = \{\mathbf{q}\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$
 216 the set of output sequences. Given a question $\mathbf{q}$, the language model defines a distribution over output
 217 sequences $\mathbf{y} \in \mathcal{Y}$ autoregressively:

$$\text{(general policy)} \quad \pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{l=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \prod_{l=1}^{|\mathbf{y}|} \text{softmax}(f_\theta(\mathbf{x}, \mathbf{y}_{<l}))_{\mathbf{y}_l}, \quad (5.2)$$

218 where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ and predicts the l -th token from the preceding context
 219 $\mathbf{y}_{<l}$, and the next token follows $\text{softmax}(\mathbf{z})_v := \exp(\mathbf{z}_v) / \sum_{v' \in \mathcal{V}} \exp(\mathbf{z}_{v'})$ with $\mathbf{z} = f_\theta(\mathbf{x}, \mathbf{y}_{<l})$.

220 Because the full expressive power of practical models renders end-to-end optimization guarantees
 221 intractable [3, 23], we follow a well-established line of theoretical work [27, 24, 10, 45] and adopt a
 222 *tabular policy* parameterization. This is the special case of Eq. 5.2 in which each output is assigned
 223 its own trainable logit indexed by the preceding token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \pi_\theta(\mathbf{y}_l|\mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

224 where $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ associated with the preceding token $\mathbf{y}_{l-1}$. This param-
 225 eterization retains the softmax structure of the general policy while making the training dynamics
 226 analytically tractable.

227 **Optimization for RL.** To analyze the training dynamics, we combine the tabular policy of Eq.5.3
 228 with the RL objective $\phi_{\text{RL}}(\theta)$ of Eq.3.1 and study the policy gradient in the small-learning-rate limit,
 229 i.e., the gradient flow

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

230 where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

231 Our experiments established that the success rate of each individual reasoning pattern stays essentially
 232 fixed throughout training. We attribute this to an inherent asymmetry in optimization difficulty:
 233 reshaping the mapping from questions to reasoning patterns is markedly easier than altering the path
 234 from reasoning patterns to final answers. We formalize this empirically grounded observation as the
 235 following assumption.

236 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 237 *$\mathbf{r}^*$, defined as $\pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

238 Throughout the remainder of this section we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$
 239 for the fixed success rate of pattern $\mathbf{r}$. We proceed in two steps: we first derive the optimal policy for
 240 the RL objective (Eq.3.1) under the general autoregressive policy (Eq.5.2), and then characterize the
 241 RLVR and RLIF training dynamics under the tabular policy (Eq. 5.3).

242 5.2 The Optimal Policy for RL

243 Our experiments showed that RLVR improves performance steadily by promoting higher-success-rate
 244 reasoning patterns, whereas RLIF yields only unstable, non-monotone gains. We now explain this
 245 contrast at its root by characterizing the optimal policy that each objective drives the model toward.

246 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 247 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

¹For simplicity, we omit the normalizing coefficient $1/|\mathcal{Y}|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 pinpoints the optimal solutions for RLVR and RLIF and, in doing so, explains their divergent behavior. Assuming π_{opt} lies within the general autoregressive policy space (which holds for both LLMs and the tabular policies analyzed below), the RLVR-optimal policy up-weights a reasoning pattern $\mathbf{r}$ precisely when its probability of deducing the correct answer is high enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$; the reward is thus *aligned with correctness*. The RLIF reward $R_{\text{RLIF}}(\mathbf{r})$, by contrast, depends only on the confidence of the final-answer distribution and cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. This single structural difference is decisive: the RLVR optimum consistently improves accuracy, whereas the RLIF optimum carries no such guarantee and may favor patterns that are confident yet wrong.

Moreover, since β is typically small in practice (e.g., 0.001) [26, 5, 44], taking the limit $\beta \rightarrow 0$ sharpens this picture into a deterministic selection rule:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit the policy collapses onto the single reasoning pattern with the largest reward $R(\mathbf{r})$, independently of how the reference model initially distributes probability across patterns. This characterizes the *destination* of optimization, but not the *path*: because the landscape is non-convex, it does not by itself reveal how reliably RLVR reaches high-reward patterns, nor why RLIF can improve performance before deteriorating. We resolve both questions in the next two subsections by analyzing the full training dynamics under the tabular policy (Eq. 5.3).

5.3 Training Dynamic Analysis for RLVR

Our experiments established that the magnitude of RLVR’s gains depends strongly on the capacity of the base model. We now explain this dependence analytically by tracking the probability assigned to the *optimal reasoning pattern*—the pattern $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$ —under gradient flow. The dynamics split cleanly into two regimes. In the first, $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically and converges to 1. In the second, the model must first pass through an *entanglement stage*, in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ absorbs probability and stalls the growth of this selection probability; only after this stage does the model transition into first-regime dynamics and ultimately converge to $\mathbf{r}^*$. The following two theorems make each regime, and the boundary between them, precise.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 isolates the favorable regime: the base reference model is strong enough that its overall accuracy already exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under this condition RLVR converges to the optimal pattern at the fast rate $\mathcal{O}(1/\epsilon)$, so the model rapidly attains an overall accuracy close to the success rate of $\mathbf{r}^*$.

The complementary, and empirically well-documented, regime arises when the base model is weaker, where RLVR optimization is known to be difficult [44, 36]. Here the model first enters an entanglement stage in which the *second-best pattern*, $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, competes for probability and slows the optimization of $\mathbf{r}^*$. Theorem 5.4 quantifies exactly how long this stage lasts.

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|q)} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|q)} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 analyzes the representative case in which only the optimal and second-best patterns have success rates exceeding the overall accuracy. We adopt this setting for analytical clarity; the argument extends, at the cost of heavier bookkeeping, to the more general situation in which *at least* two patterns exceed the average success rate.

The central consequence of Theorem 5.4 is a sharp separation of time scales. Reaching the favorable regime of Theorem 5.3—where the overall accuracy surpasses the success rate of the second-best pattern and the optimal pattern is selected with substantial probability—requires T_0 steps. Whereas the subsequent convergence time T_1 in Theorem 5.3 is merely polynomial in $1/\epsilon$, the transition time T_0 can grow *super-exponentially* in $\gamma_{\pi_{\text{ref}}}$, the ratio between the total selection-weighted success mass carried by $\mathcal{R}/\{\mathbf{r}'\}$ and that carried by $\mathbf{r}^*$. When the base model places only a vanishing probability on the optimal pattern, $\gamma_{\pi_{\text{ref}}}$ becomes large and T_0 prohibitive, so the model can remain stuck for an extended period before it begins to select $\mathbf{r}^*$ with appreciable probability—precisely the entanglement stage. This result gives a rigorous explanation for why RLVR succeeds effortlessly on strong base models yet stalls on weak ones.

We corroborate both regimes with case studies and numerical simulations in Section 6.

5.4 Theoretical Explanation for RLIF

Proposition 5.2 already shows that RLIF, being driven solely by the confidence of the final-answer distribution, cannot distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives, and thus that its optimum may lie below the base model in accuracy—consistent with the degradation we observe empirically. Yet prior work reports that RLIF can improve performance without any external reward Agarwal et al. [4], Zhao et al. [47]. We reconcile these seemingly opposing facts by analyzing RLIF’s dynamics, showing that both the early improvement and the eventual collapse are predicted by the same theory.

We begin with the early stage, considering a well-trained base model that satisfies the following assumption.

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|q, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|q) = \mathbf{a}^*$.*

Because majority voting [35] is a proven means of improving model performance, a well-trained LLM typically assigns the correct answer $\mathbf{a}^*$ higher probability than any incorrect one, so Assumption 5.5 is well justified for modern LLMs. Under this assumption we analyze the accuracy dynamics at initialization ($t = 0$) in the case $|\mathcal{A}| = 2$, with per-pattern success rates drawn uniformly and a high-entropy pattern-selection distribution², obtaining the following theorem.

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

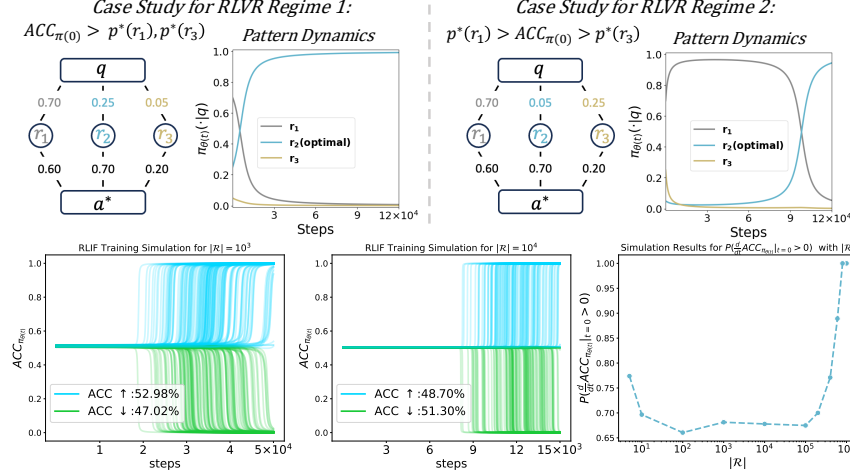


Figure 2: Empirical validation of the theory. *Upper (RLVR case studies)*: the two regimes of Theorems 5.3 and 5.4—rapid convergence to the optimal pattern (left) and the entanglement stage that delays it (right). *Bottom (RLIF simulations)*: across varying $|\mathcal{R}|$, the probability of converging to an improved versus a degraded state (left) and the probability of an initial performance gain (right). The simulations reproduce both predictions of Theorem 5.6: near-certain early improvement together with a near one-half chance of eventual degradation.

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt}ACC_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The two results of Theorem 5.6 together resolve the apparent paradox around RLIF. The first shows that the accuracy derivative at initialization is positive, which explains RLIF’s early performance gains. The second exposes the failure mode that follows: as training proceeds, Proposition 5.2 implies that at $\beta = 0$ the policy converges deterministically to the pattern maximizing $R_{\text{RLIF}}(\mathbf{r})$, and with probability 0.5 this maximizer coincides with the *least* accurate pattern, $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Thus RLIF improves the model at the outset yet, with probability one-half, ultimately converges toward the lowest-accuracy reasoning path—driving performance below that of the base model. This rise-then-collapse trajectory is exactly the behavior we characterize theoretically here and observe empirically, now derived from first principles.

6 Case Studies and Numerical Simulations

To validate and concretely illustrate the theory, we present case studies and training simulations that instantiate each theorem. For RLVR, we realize the two regimes of Theorems 5.3 and 5.4, shown in Fig. 2 (upper). For RLIF, we run repeated training simulations in which each reasoning path’s success rate is drawn uniformly, directly probing RLIF’s behavior both at the initial training steps and at convergence, shown in Fig. 2 (bottom). In every case the simulated dynamics match the theoretical predictions.

6.1 Case Studies for RLVR

Section 5.3 established two regimes in RLVR’s training dynamics: in the first (Eq 5.7), the probability $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ of selecting the optimal pattern rises monotonically to convergence; in the second (Eq 5.8), an initial entanglement phase forces a substantially longer training period before the first-regime dynamics take over and the model converges to $\mathbf{r}^*$. The case studies confirm both:

- When the reference model satisfies Eq 5.7—its overall accuracy exceeding the success rate of every non-optimal pattern—the model converges rapidly to the optimal reasoning pattern, as shown in Fig. 2 (upper left).
- When instead a suboptimal pattern $\mathbf{r}'$ has a higher success rate than the initial overall accuracy (Theorem 5.4), the model first traverses the entanglement phase, and a transition period T_0 (Theorem 5.4) must elapse before the rapid first-regime optimization begins. Fig. 2 (upper right) instantiates a large- $\gamma_{\pi_{\text{ref}}}$ scenario (with $\gamma_{\pi_{\text{ref}}}$ defined in Eq 5.9, here $\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$);

because T_0 scales as $\gamma^{\pi_{\text{ref}}}$, the transition becomes prohibitively long and convergence to the optimal pattern is markedly delayed. The observed pattern dynamics exhibit exactly this entanglement stage, in close agreement with the theory.

6.2 Numerical Simulations For RLIF

Section 5.4 established that, under Assumption 5.5 and the base-model distribution of Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the probability of converging to the least accurate reasoning path is 0.5. Our simulations confirm both predictions. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$: the probability of converging to an improved state ($ACC \uparrow$) closely tracks that of converging to a degraded state ($ACC \downarrow$), matching the second result of Theorem 5.6. Fig. 2 (bottom right) further shows that, at the initial training step, the probability of performance improvement approaches 1 as $|\mathcal{R}|$ grows, matching the first result. The simulations thus reproduce both the early gain and the eventual coin-flip between improvement and degradation predicted by the theory.

7 Conclusions and Limitations

This work delivers a unified account of reinforcement learning dynamics in LLMs, coupling systematic empirical analysis with a rigorous theoretical framework. Viewing RL through the lens of reasoning-pattern selection, we show that RL reshapes the distribution over reasoning patterns via sparse, token-level interventions while leaving each pattern’s intrinsic success rate intact, and we translate this observation into provable training dynamics for the two dominant reward mechanisms: for RLVR we characterize the fast-convergence and entanglement regimes and identify the base model’s reasoning quality as the decisive factor, while for RLIF we explain both its early gains and its eventual degradation. Case studies and simulations confirm every prediction. Our analysis rests on deliberate simplifications that also mark the natural directions for future work: the interpretability of LLM-identified reasoning patterns merits further validation, the two-stage framework can be extended to richer real-world reasoning settings, and the dynamics results can be generalized beyond the specific base-model assumptions adopted here. We view these as opportunities to broaden a framework whose current form already accounts for the full range of RL training behaviors observed in our experiments.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 analyzed the reasoning-pattern dynamics on Number Theory tasks from MATH. To confirm that this behavior is not task-specific, we repeat the dynamic analysis on Geometry tasks (Figure 3). The same contrast recurs: under RLVR the model steadily concentrates on the higher-accuracy patterns, whereas under RLIF pattern selection remains unstable. This replication is fully consistent with the empirical and theoretical results of the main paper.

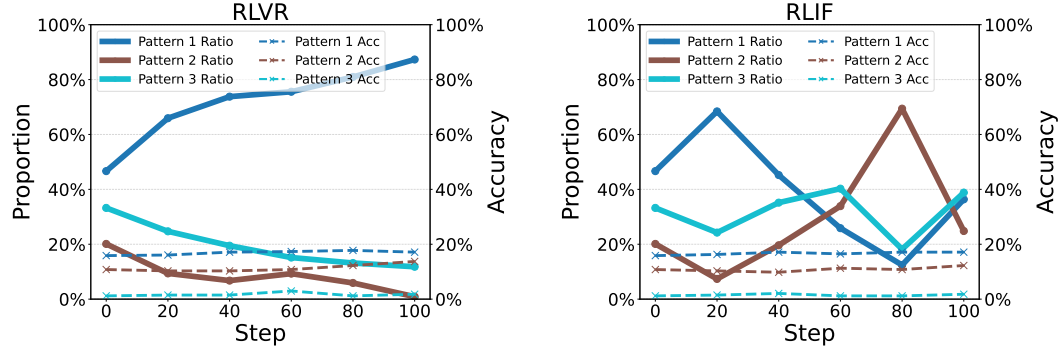


Figure 3: Reasoning-pattern distribution and corresponding accuracy of RL-enhanced models on Geometry tasks. Under RLVR the model converges toward the highest-accuracy pattern (Pattern 1), whereas under RLIF the pattern distribution remains unstable—mirroring the Number Theory dynamics in the main paper.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To establish that our findings hold across model families rather than a single base model, we conduct supplementary experiments validating both the empirical insights and the theoretical conclusions (Figure 4). Here we take Qwen-2.5-7B-Instruct [39] as the base model and compare it against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. Our evaluation spans a broad range of mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—using challenging problems sampled from the MATH dataset [16], and we further extend it to complex problems from AMC23 [1]. Across every domain the results reinforce our earlier findings: RLVR increases the frequency of high-accuracy reasoning patterns and suppresses less effective ones. Task-specific analyses are provided in Appendix B.3.

Across these tasks, the highest-success-rate patterns consistently become more prevalent after RLVR enhancement, reinforcing the findings of Section 4 and Section 5. Notably, in tasks such as Algebra and AMC23-19 (Figure 4), the per-pattern success rates remain stable across the base and RL-enhanced models; this not only echoes our earlier observations but also provides direct empirical support for Assumption 5.1 underlying our theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To demonstrate that our theory scales to substantially larger models, we further experiment on QwQ-32B [32], which is built on Qwen-2.5-32B-Instruct [39] and markedly strengthens reasoning through RLVR. We measure the reasoning-pattern shift between these two models on four tasks: number theory and geometry tasks from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

The result is unambiguous: the reasoning patterns of QwQ-32B predominantly coincide with the most accurate patterns of Qwen-2.5-32B, exactly as our theory predicts. We omit per-pattern accuracies for QwQ because API-only access prevents us from extracting the CoT data needed to have the base model continue from a fixed reasoning prefix, which is required to attribute accuracy to a reasoning pattern rather than to the model as a whole; see the “Accuracy Analysis” in Appendix B.2 for details.

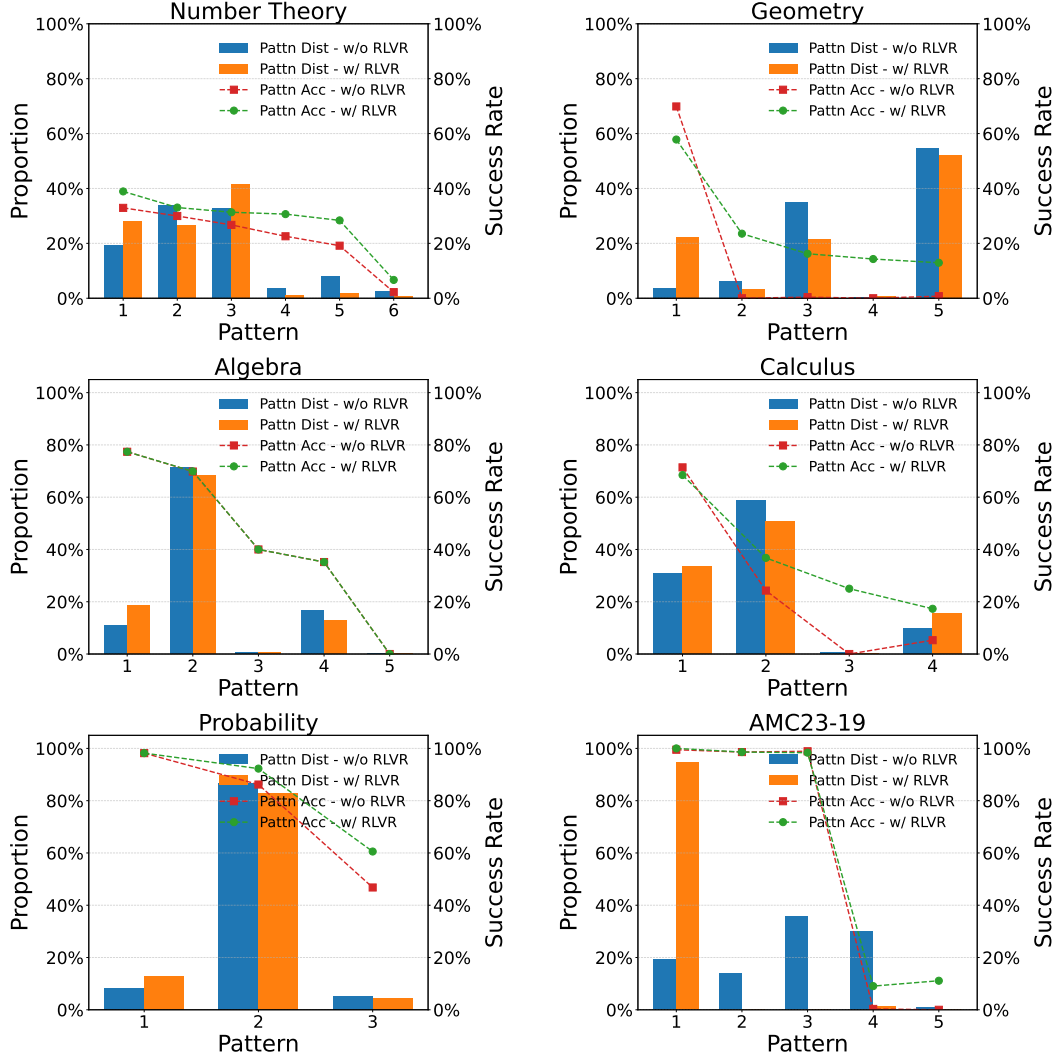


Figure 4: Reasoning-pattern distributions and corresponding success rates, with and without RLVR enhancement, across additional tasks; the results align closely with our experimental insights and theoretical conclusions. Bars (Pattern Dist) give each pattern’s proportion among all patterns, and dots (Pattern Acc) give its corresponding accuracy or success rate. In every task, RLVR shifts probability mass toward the highest-accuracy patterns.

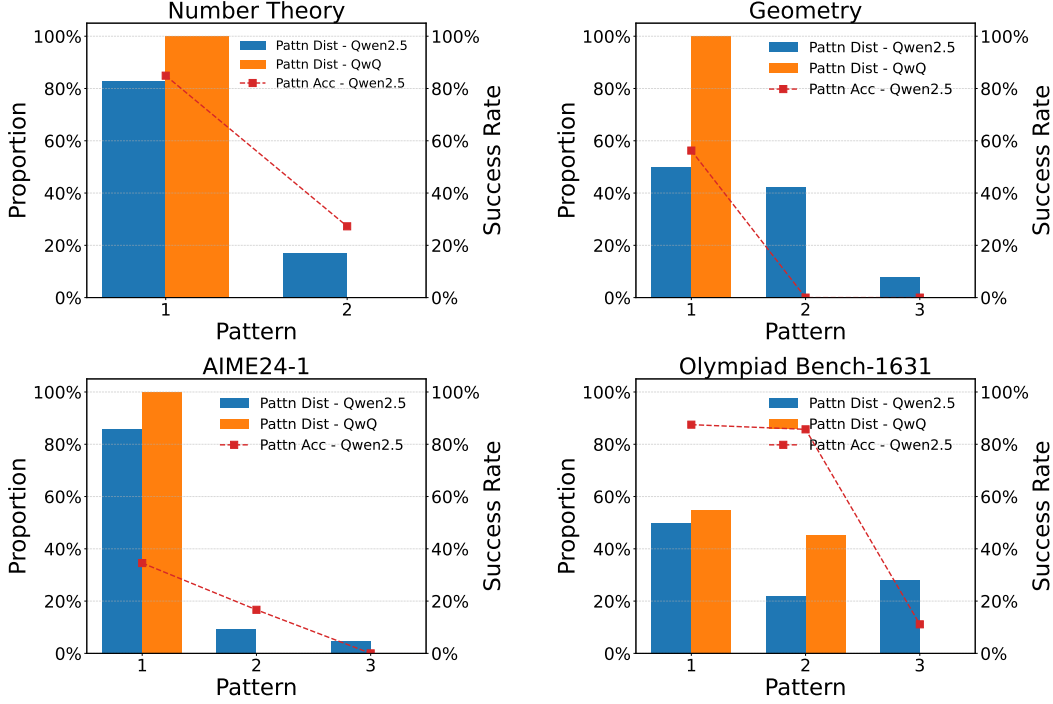


Figure 5: Evaluation on the larger models Qwen-2.5-32B and QwQ-32B. QwQ’s reasoning patterns converge onto the most accurate pattern of Qwen-2.5, demonstrating that our theory extends to large-scale models.

B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We take Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. All runs use the verl [31] framework with parameter settings identical to verl’s provided MATH example, and RLIF differs from RLVR solely in the reward computation, isolating the reward as the only source of variation. For each method we obtain three independent training curves by varying the random seed.

Token-level Analysis: This experiment uses Qwen2.5-3B together with the RL-enhanced models above. We randomly sample 400 questions from each of the GSM8K and MATH test sets and use the full AIME24 dataset. For each question we first sample an answer from the base model, then concatenate this answer to the question and feed the result to the RL-enhanced model as a new prompt. Using the OpenAI library interface, we compute, for every token in the prompt, its probability and its rank among the probabilities of all candidate tokens at that position. Comparing each token’s rank before and after RL yields the proportion of tokens whose rank changed, which quantifies how sparsely RL intervenes at the token level.

Reasoning-pattern Level Analysis: We conduct two complementary experiments. First, to trace the pattern distribution *during* training, we use our own models—Qwen2.5-3B and its RL-enhanced counterpart—and evaluate every 20 steps. Second, to test the generality of our conclusions across different and larger base models, we evaluate the open-source Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B against their respective base models; for these, only the pre- versus post-RL pattern distributions are accessible, so we examine the induced distribution shift rather than the full within-training dynamics.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We randomly sample questions spanning several task types (e.g., Geometry, Algebra) and difficulty levels (Level 4, Level 5) from datasets including MATH. For the within-training pattern-dynamics experiment, the model samples 1024 answers per question; for the experiment that considers only the pre- versus post-RL distribution change, it samples between 64 and 1024 answers per question, depending on the task category. These responses serve as the input to the subsequent pattern extraction and classification stages.

Reasoning Patterns Extraction: From a sampled subset of base-model responses, we use the GPT-4o [19] API to summarize the pattern categories; the full extraction prompt appears in Appendix D.1. To make extraction deterministic and reproducible, we set the GPT-4o temperature to 0.

Responses Classification: We again employ GPT-4o, at temperature 0, to assign every response to one of the extracted pattern categories; the full classification prompt is given in Appendix D.2.

Accuracy Analysis: To isolate the effect of the pattern distribution on accuracy from any change in the reasoning process itself, we hold the reasoning process fixed and vary only the completion. Concretely, we take a base-model response, remove the sentence containing the final answer, append the remaining prefix to the question, and use this as a new prompt from which the RL-enhanced model continues to predict the answer. Since the partial response fixes a specific reasoning pattern, the accuracy of the model’s continuation measures the success rate of that reasoning pattern rather than of the model as a whole, letting us attribute accuracy changes solely to the pattern distribution.

B.3 Details for Each Task

Below we detail the experimental procedure for each task, including task descriptions and representative reasoning-pattern examples (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: This task poses coin-distribution problems over multiple bags: the bags initially hold equal numbers of coins, and after additional coins are received and redistributed so that every bag again holds the same count exceeding a specified total, the model must determine the minimum initial number of coins per bag. We adapt this task from a level-5 MATH problem (id: test/number_theory/1055) [16], generating 32 variants by adjusting parameters such as the bag count and coin totals to form the evaluation set. We use one-shot prompting with a simple, unrelated example that fixes the answer format via "boxed" notation without biasing the reasoning approach. For evaluation, we extract reasoning patterns from 20 questions with 4 responses each, sample 64 responses per question, and then classify all responses by the extracted patterns, enabling a direct comparison of the reasoning-pattern distribution before and after RL.

Geometry: In the geometry task, the model is given the lengths of two sides of a scalene triangle and asked how many distinct integer values (in centimeters) the third side can take. This task is adapted from a level-4 MATH geometry problem (id: test/geometry/1046). Its evaluation follows the same pipeline as Number Theory: we build a synthetic set of 32 questions, extract reasoning patterns from 20 questions with 4 base-model (Qwen2.5-7b-Instruct) responses each, and then compare the model with and without RLVR enhancement across the full evaluation set at 64 responses per question.

Algebra: The algebra task is a can-recycling problem: starting from a given number of old cans, and given that n old cans can be recycled into one new can, the model must determine how many cans can ultimately be produced. The difficulty lies in the iterative accounting—at each recycling step some cans cannot be divided evenly, and these leftovers may later be combined for further recycling. The template is adapted from a level-4 MATH algebra problem (id: test/algebra/2768). For this and all subsequent supplementary tasks, we adopt exactly the same settings as the earlier tasks, including dataset size and number of samples.

Calculus: The calculus task is deliberately compact: the model is given a single-variable trigonometric equation together with the domain of the variable and must determine the number of solutions.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning-pattern examples across the evaluation tasks, with their defining keywords and common elements.

Though simple, the task probes the model’s basic competence on calculus problems. Its prototype is a level-4 MATH pre-calculus problem (id: test/precalculus/1140).

Counting and Probability: We further evaluate the model on a counting-and-probability task involving white and black balls. Given the probability that, when two balls are drawn at random, one is black and one is white, together with the number of balls of one color, the model must compute the minimum number of balls of the other color. This probability task is adapted from a level-4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

AMC23 Question: For this task we use question 19 of AMC23 [1]: we randomly sample 48 responses for reasoning-pattern extraction and 1024 responses for the subsequent pattern analysis.

B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

We evaluate four tasks. Because API access constrains testing, we select one question per task, sample 64 responses from each of the two models, and perform reasoning-pattern extraction and classification directly on these responses. Specifically, from the MATH dataset we use question 1055 (number theory) and question 1046 (geometry), and for the remaining two tasks we use problem 1 from AIME24 and question 1631 from OlympiadBench. Representative patterns are shown in Table 2.

665 C Deferred Proofs

666 This appendix presents the complete proofs of our main theoretical results: Proposition 5.2 and
667 Theorems 5.3, 5.4, and 5.6.

668 C.1 Proof of Proposition 5.2

669 *Proof of Proposition 5.2.* In this proof, we utilize the proof techniques in [26], recall that the opti-
670 mization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

671 Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample
672 $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$
673 can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{r}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

674 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

675 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the
676 partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

677 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

678 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

679 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
680 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

681 specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log (\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log (|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|\mathbf{q})) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

682 where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned}\pi_{opt}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{ref}}(\mathbf{r}_i|\mathbf{q})}\end{aligned}$$

683 So we can write $R_{RLIF}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
684 of the theorem. $\square$

685 C.2 Dynamics of Tabular Policy

686 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
687 gradient dynamics of θ can be computed as

$$\begin{aligned}\frac{d}{dt} \theta(t) &= \nabla \phi_{RL}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})],\end{aligned}$$

688 Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to $\mathbf{q}$, and
689 $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned}\frac{\partial \theta_{r_i,q}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})\end{aligned}$$

690 where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we
691 have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t}$$

692 C.3 Proof of Theorem 5.3

693 The proof of Theorem 5.3 proceeds in three steps: we first show that the overall accuracy is non-
694 decreasing, $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then establish a lower bound on $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$
695 (Eq C.5); and we finally derive the convergence-time bound on t stated in the theorem.

696 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{r_i,q}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

697 For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned}\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j,q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j,q}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}),\end{aligned}$$

698 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\
&\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \tag{C.4}
\end{aligned}$$

699 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
700 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
701 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

702 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
703 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

704 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

705 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

706 This concludes the proof of the theorem. □

707 C.4 Proof of Theorem 5.4

708 The proof of Theorem 5.4 proceeds as follows: we first establish an upper bound on $\pi_{\theta(t)}(\mathbf{r}' | \mathbf{q})$
709 (Eq C.6), then analyze the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})}$ (Eq C.7), and thereby prove that $\text{ACC}_{\theta(t)} \geq$
710 $p^*(\mathbf{r}')$ for all $t > T_0$.

711 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 712 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

713 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

714 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

715 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 716 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

717 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

718 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

719 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

720 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned}
\text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\
&= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\
&\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right].
\end{aligned}$$

721 Let $C_2 = 1/\Delta, C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

722 then:

$$\begin{aligned}
&\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
&\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
&\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\
&= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\
&= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}').
\end{aligned}$$

723 Finally, for $t > T_0$ we have:

$$\begin{aligned}
\text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\
&\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0.
\end{aligned}$$

724 This concludes the proof of the theorem. $\square$

725 C.5 Proof of Theorem 5.6

726 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
727 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\
&= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right),
\end{aligned}$$

728 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 729 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

730 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 731 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 732 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 733 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

734 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 735 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 736 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 737 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

738 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 739 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

740 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 741 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

742 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

743 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 744 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 745 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 746 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 747 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 748 $1 - 6\delta$.
- 749 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 750 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

751 This concludes the proof of the second statement.

752

□

753 D Full Prompts

754 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

755

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

756

757 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

758

759 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

760

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

761

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

762

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

763

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

764

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

765

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

766

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

767

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

768

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a s -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

769

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

770